

Full Test — Answers

Final answers with a one-line reason. For full methods, revisit the concept notes.

Objective

Q1 (c) **50**. $L = 50$.

Q2 (b) **XLIX**. $40 (XL) + 9 (IX)$.

Q3 (b) **powers of n**. Each landmark is the previous $\times n$.

Q4 (c) **700**. The 7 is in the hundreds place.

Q5 **1033**. $1 \times 125 + 0 \times 25 + 3 \times 5 + 3 \times 1 = 143$.

Q6 **True**. Roman numerals have no zero.

Q7 (a) **base-10**. Landmark numbers 1, 10, 100, 1000...

Q8 (a) **Both true, R explains A**. The placeholder role is why zero is essential.

Short answer

Q9 **MMCCCLXVII**. $2367 = 2000 + 300 + 60 + 7$.

Q10 **2244 (base 5)**. $324 = 2 \times 125 + 2 \times 25 + 4 \times 5 + 4$.

Q11 $(4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$.

Q12 **8000**. The 8 is in the thousands place.

Long answer

Q13 **143**. $1 \times 125 + 0 \times 25 + 3 \times 5 + 3 \times 1 = 125 + 15 + 3$.

Q14 (i) **CDXLIV** (ii) **MCMLXXXIV**. $444 = CD + XL + IV$; $1984 = M + CM + LXXX + IV$.

Q15 **Place value 6000, face value 6, difference 5994**.

Q16 **Any three**: only ten symbols; place value; a symbol for zero; and easy arithmetic (especially multiplication).

Total: 28 marks.